ECE320/L Quartus Tutorial: Half Adder Design GLaw

Schematic Design Entry Method with Modelsim Simulation

Notation:

To simplify what one should click to perform a task, the following form of notation will be used:

Menu item → Menu item → … → Menu item

Example:

File → Project Wizard means "on the File menu, click New Project Wizard"

Objective

To learn to use Quartus to develop and simulate a half adder using Schematic Design Entry Method

Development Tools

Quartus 11.0

Hardware

DE2 board, USB Blaster, and USB cable. (Will try them out in ECE420)

References

Altera Quartus Handbooks Volume 1, 2, and 3 available at www.altera.com or at the course website www.csun.edu/~glaw

Launch Quartus II Software

1. Launch the Quartus II software.
2. File → Project Wizard (on the File menu, click New Project Wizard)
3. New Project Wizard: Introduction windows appears. It shows you the five steps you may need to go through in setting up a new project. You have nothing to do in this window, hence click **Next**.
4. New Project Wizard: Directory, Name, Top-Level Entity window appears.

   a. Enter the working project directory in your Z drive, for example, z:\...\Tutorial1

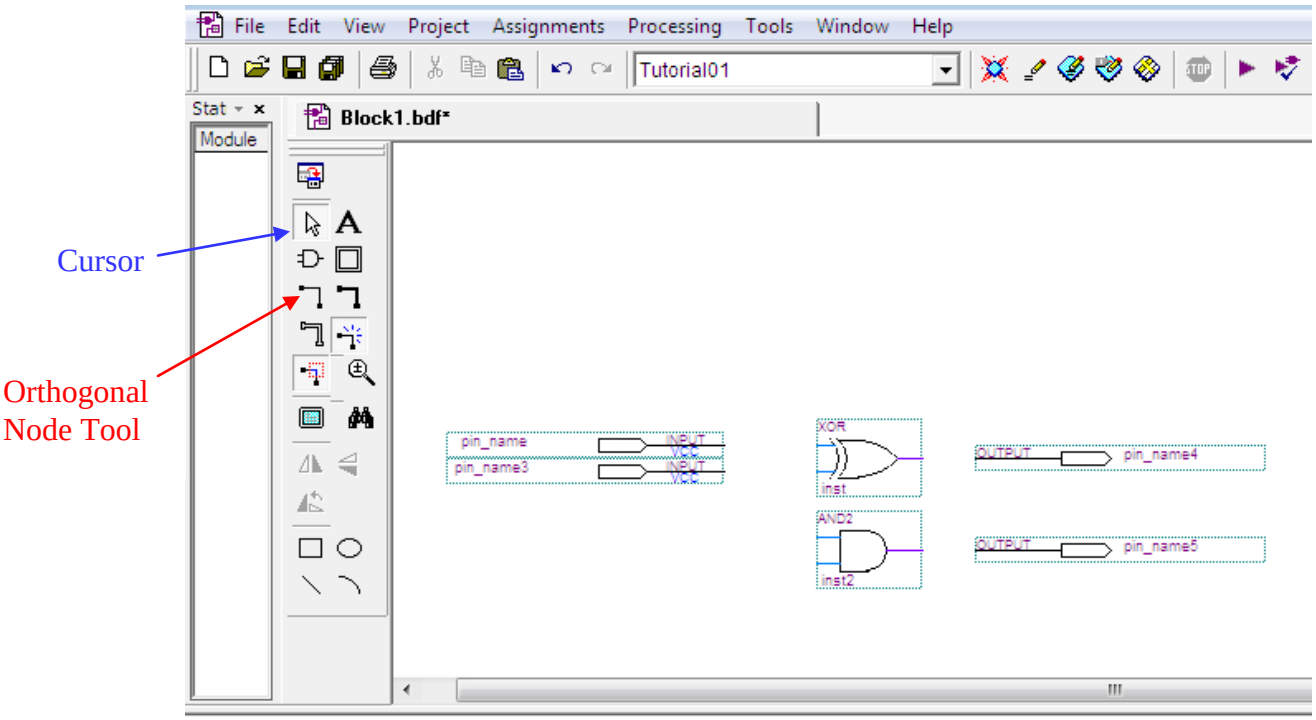
   b. Enter project name: HalfAdder. As you type HalfAdder name, Top-Level Entity name will be automatically filled with HalfAdder.

   c. Click next. When asked to create z:\...\Tutorial1 directory, click **Ye**s.
5. New Project Wizard: Add Files window appears. You have no files to add now. Hence, just click **Next**.

6. New Project Wizard: Family & Device Settings window appears.
   Scroll up/down the family choices and choose Cyclone II family. Scroll up/down the available devices and choose EP2C35F672C6. Leave other options as they are. Click **Finish**.

Create a Half Adder

1. File → New. New file option window appears.
2. Select Block Diagram/ Schematic File. Click OK. BDF Editor window appears.
3. Double click anywhere on the window. Select Symbol window appears.
4. Type xor in the Name box. Click Ok. Drag the XOR symbol and put it in the middle of the window.
5. Repeat steps 3 and 4 but with and2 symbol name to put a 2-input AND symbol at the bottom of the XOR symbol.
6. Input ports and output ports are symbols with names "input" and "output", respectively. Thus, repeat steps 3 and 4 with symbol name "input" and "output" to put two input ports and two output ports on the window. Click and drag the symbols on the window to match the BDF diagram shown below.



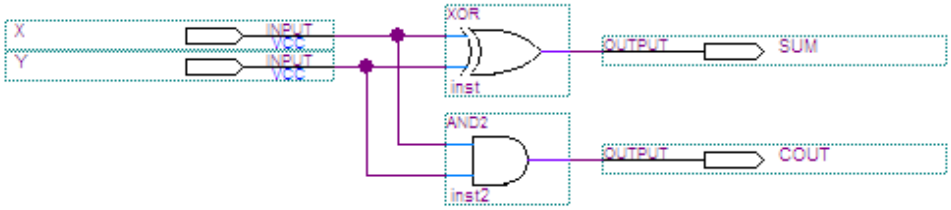
7. Click the orthogonal node tool in the list of tool on the left of the BDF window (see above). To connect point A to point B, position the cursor on point A, click and hold the left button of the mouse, drag the line and drop it at point B.

   Example: To connect input symbol to the XOR symbol input, position the cursor on the right end point of the input symbol, click and hold the left mouse button, drag the line and drop it on the input of the XOR symbol.

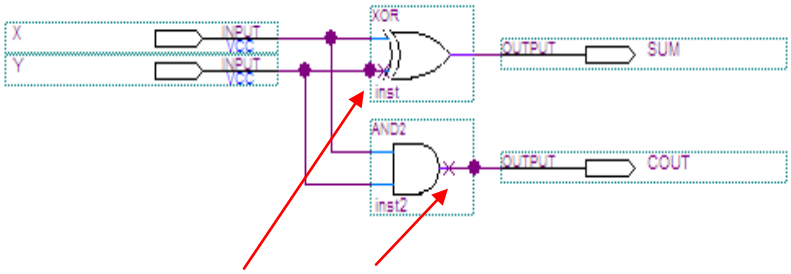
   Follow the steps above connect the input symbols and the output symbols to the XOR and AND symbols.

8. Click the cursor in the list of tool on the left of the BDF window (see above). Double click the input symbol and change the input name to X. Double click the other input symbol and change the input name to Y. Repeat the steps to name the output names to SUM and COUT, respectively.

   Now, your bdf file must look similar to the one shown below:



   To help you in identifying erroneous line connection, an example is shown below:



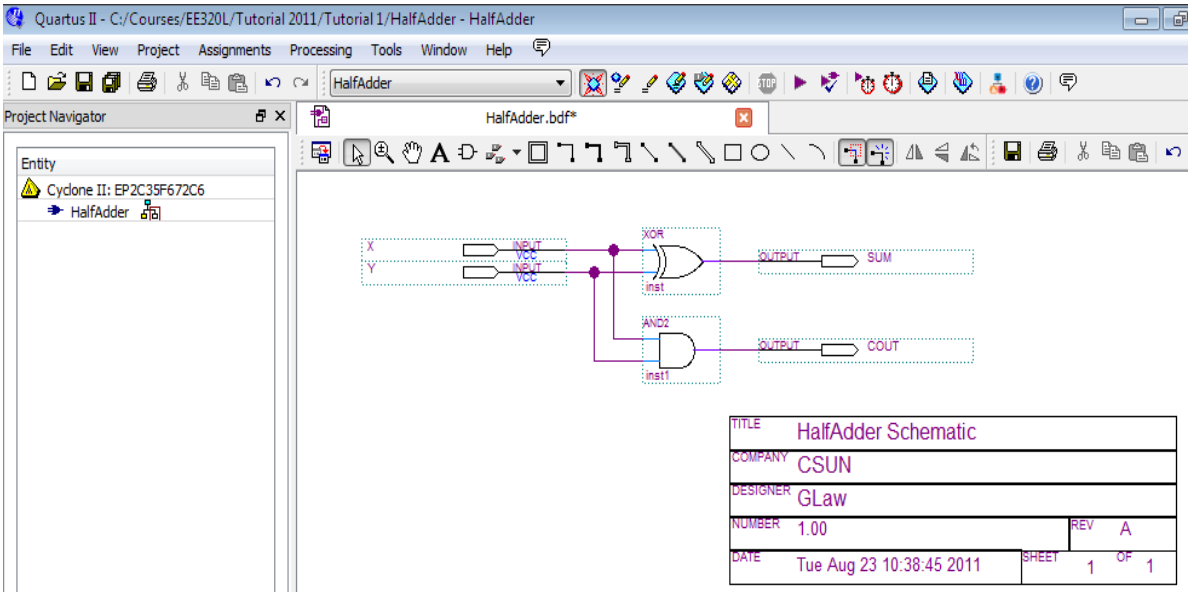
   The errors could be identified as “x” on the lines. If you see these “x”s, delete the lines and redraw them.

9. Double click anywhere on the window. Select Symbol window appears.

10. Type title in the Name box. Click Ok. Drag the title symbol and put it the lower right-hand part of the window

11. Double click the title symbol, the title block properties window appears. Click any field that you want to modify and change the value. As an example, click on Designer and change the value to your name.

12. Your final design should be similar to the one shown below:

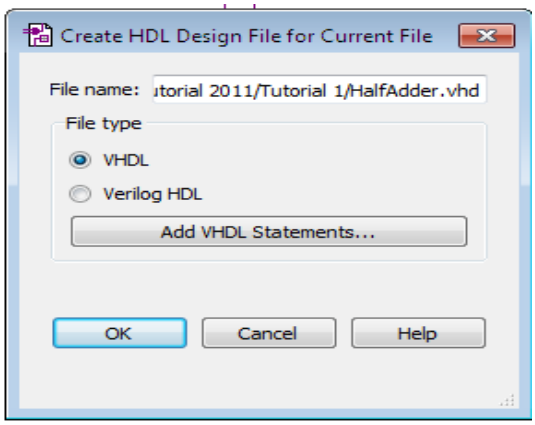


13. Save the bdf file as HalfAdder.bdf, the same name you used as the project name with bdf extension.

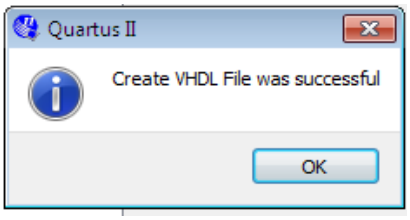
Note: Starting from Quartus version 10, Altera no longer includes the waveform-based simulator in Quartus. To do the simulation, Altera provides the third party Mentor Graphics simulator "Modelsim" instead. Hence, we need to set up Quartus so that it would generate the files necessary for Modelsim to simulate the design.

Convert BDF File to VHDL File

1. File → Create/Update → Create HDL Design File from Current File. "Create HDL Design File from Current File" window appears.
2. Choose VHDL file type and use the default file name HalfAdder.vhd. Click OK.

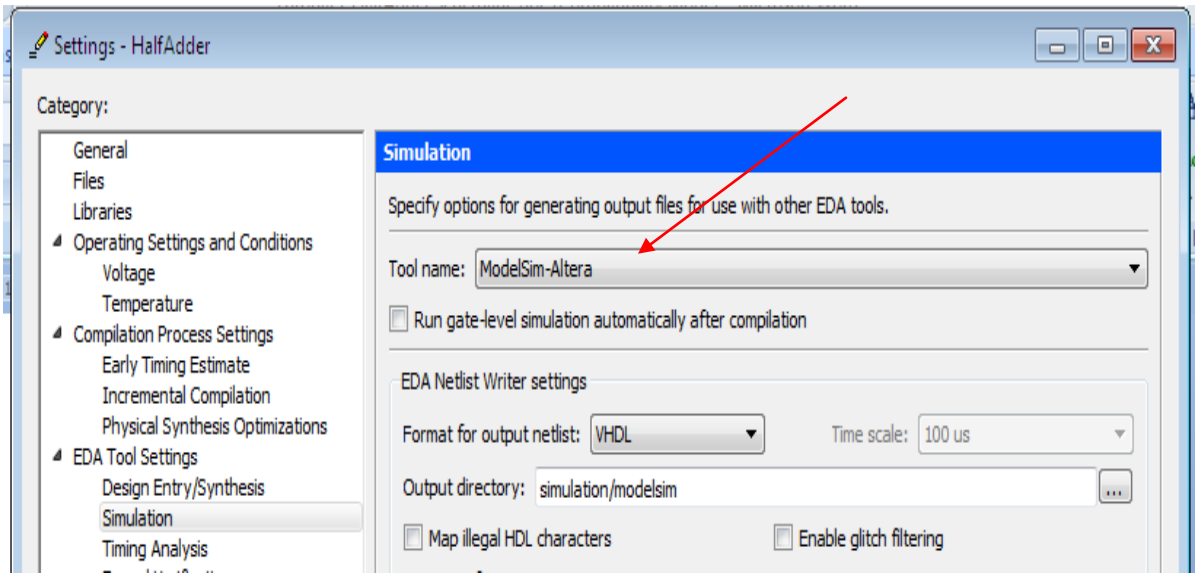


3. After the HalfAdder.vhd has been successfully created, click OK.

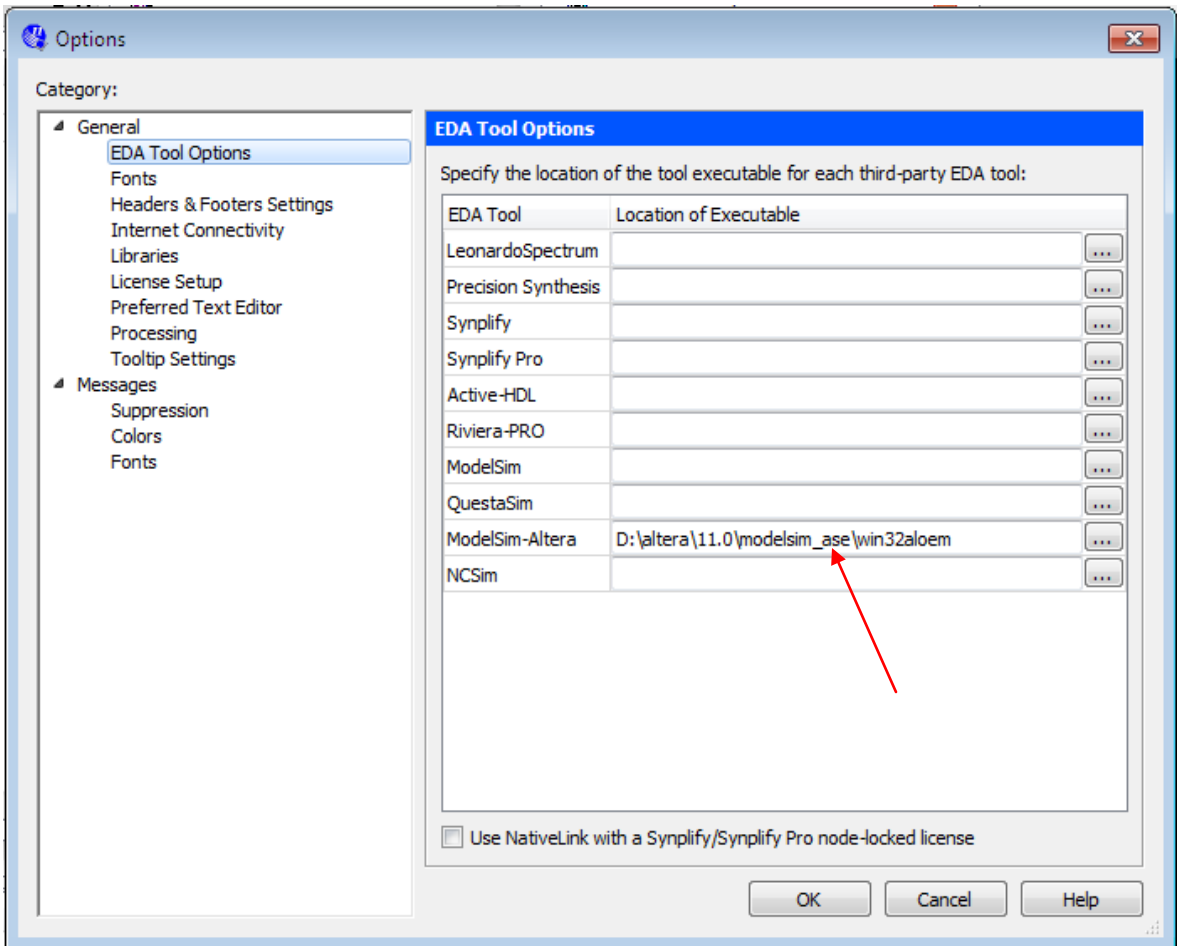


Set Quartus to Use ModelSim Simulator

1. Assignments → Settings. HalfAdder settings windows appears.
2. Under EDA Tool setting, click simulation. Browse Tool Name and choose Modelsim-Altera. Leave the other fields at the default values and click OK.



3. Tools → Options. Tools Option window appears. Under Category /General, select EDA Tool Options. In the Modelsim-Altera field, browse the folders and locate win32aloem folder for the Modelsim executable file. Click OK.



Compile the half adder

Processing → Start Compilation. If there are any errors in the message window, fix the errors and compile the design again. For this tutorial, you may ignore the warnings.

You may leave Quartus open or minimize it. As in any design, the first simulation may not give the correct result, and you may need to use Quartus again to correct the design flaw.

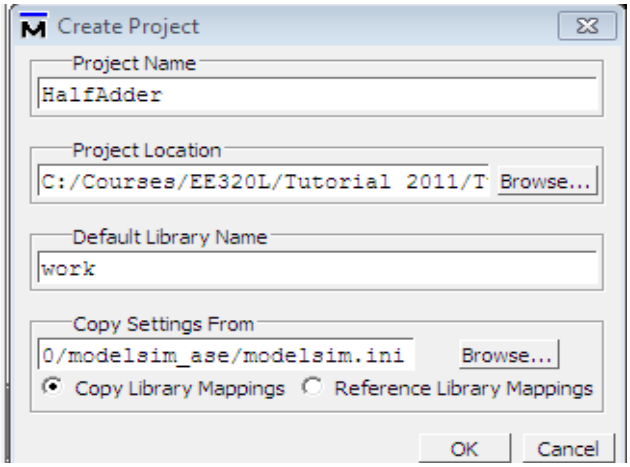
Simulate the half adder (Functional Simulation)

Note: Since you have not learned VHDL yet (ECE420), you are bound to simulate the design using waveform vectors instead of VHDL Testbench. As a note, simulating a design using VHDL Testbench is the standard procedure in the industry.

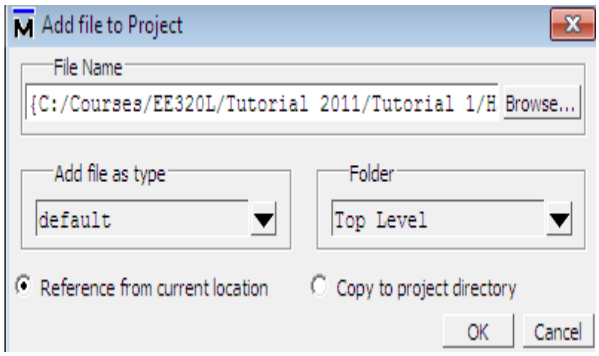
1. Launch Modelsim. When the welcome to version 6.6d window appears, click jumpstart.

2. When Welcome to Modelsim appears, click Create a Project. You can also create a new project later by File → New → Project.

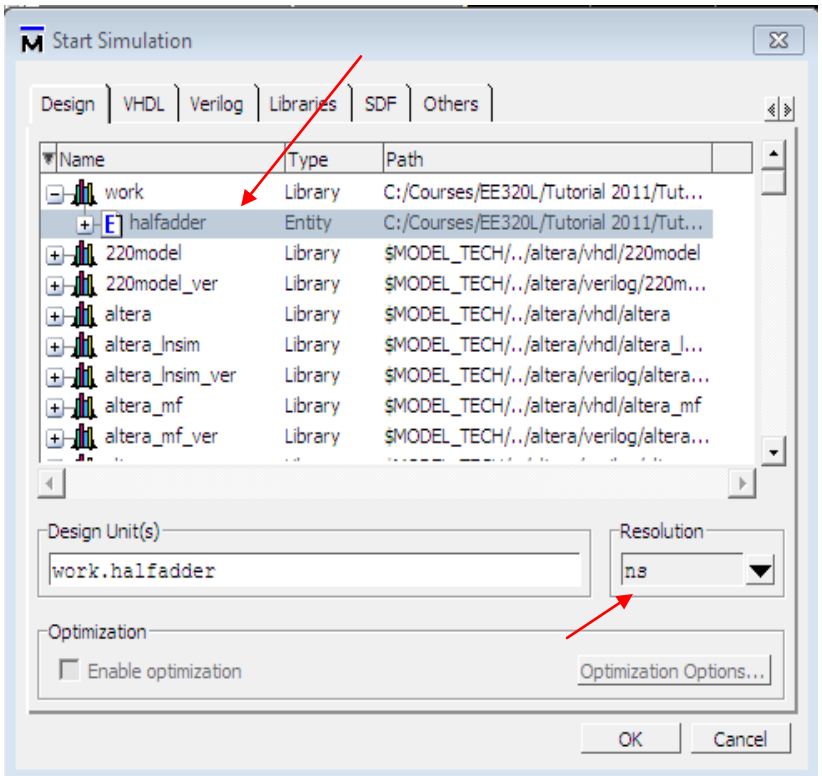
3. Enter the Project name "Halfadder" and project location "z:\...\Tutorial1" (the same project location specified in Quartus earlier). You may also use the browse feature to select the folder. Click OK.



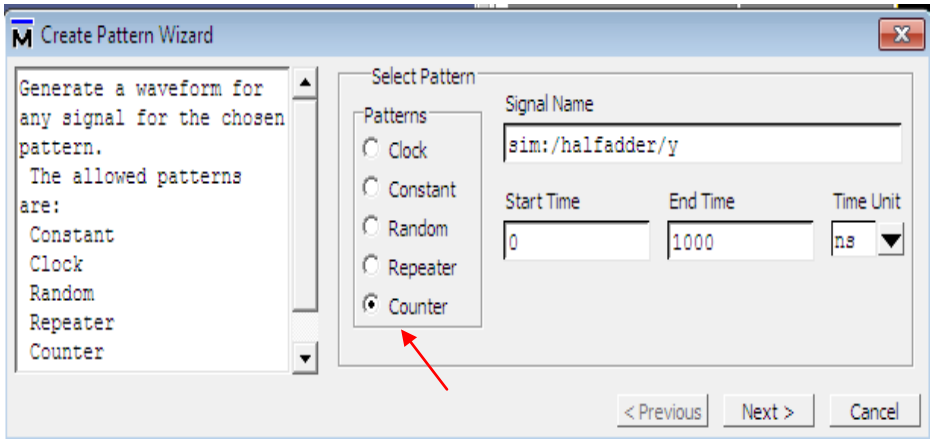
4. When Add Item to Project window appears, click Add Existing File. Browse to HalfAdder.vhd file created in Quartus. Click OK and close Add item to Project window.



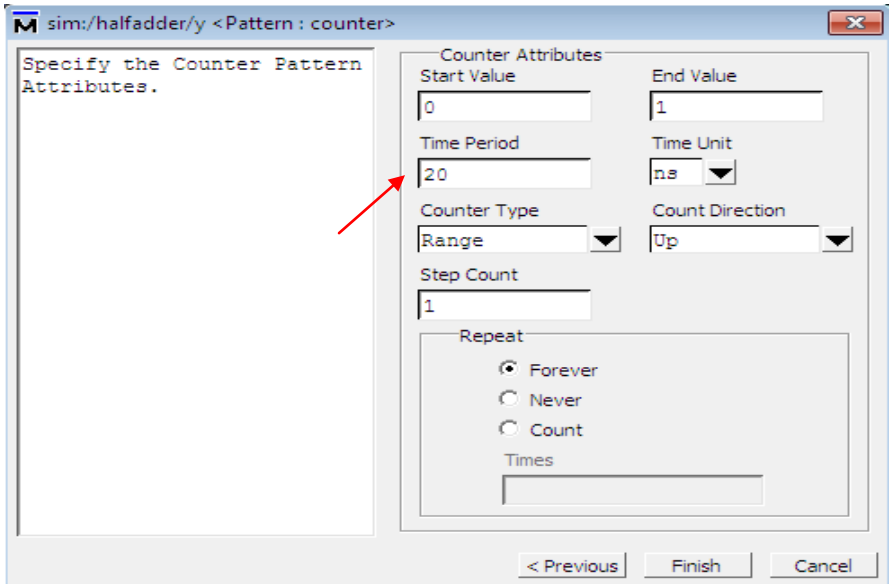
5. *Select HalfAdder*.vhd in Project pane. Compile → Selected File. After the compilation, the Compile of HalfAdder.vhd was successful message should appear in the transcript (message) pane.
6. Simulation → Start Simulation. In Start Simulation window, expand work library and select HalfAdder. Choose ns resolution. Click OK.



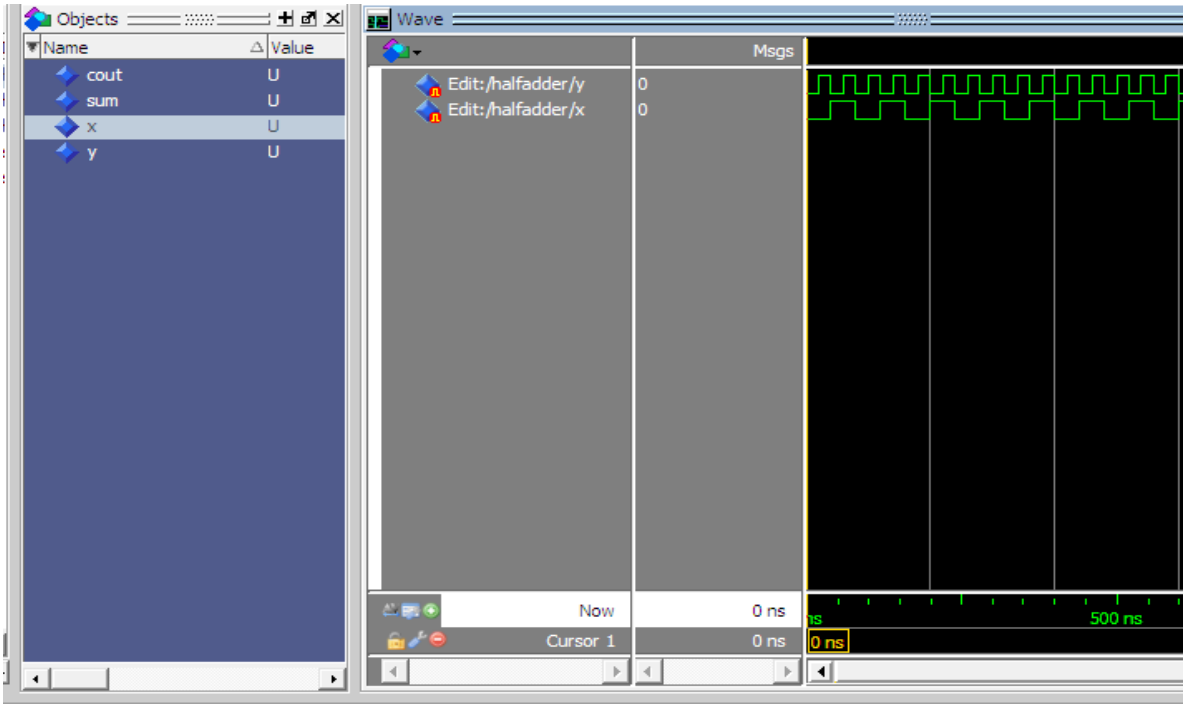
7. View →Object and View Wave. To view Objects and waveforms.
8. Right click on y signal in Object pane and click Create Wave. In Create Pattern Window, select Counter pattern. Click Next.



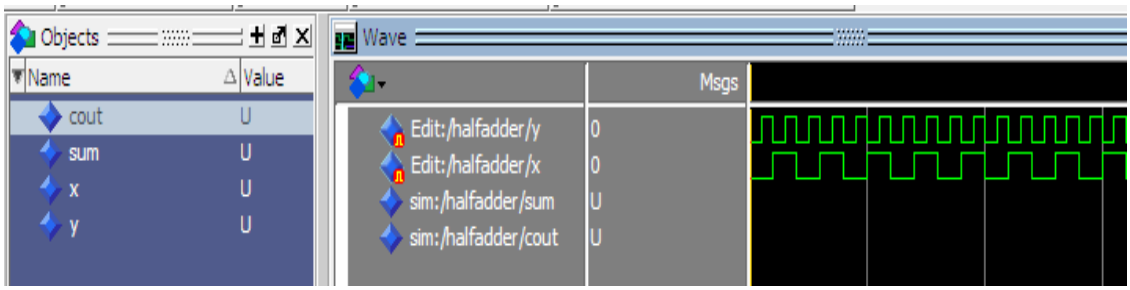
9. In the next window, set Period to 20 ns. Click Finish.



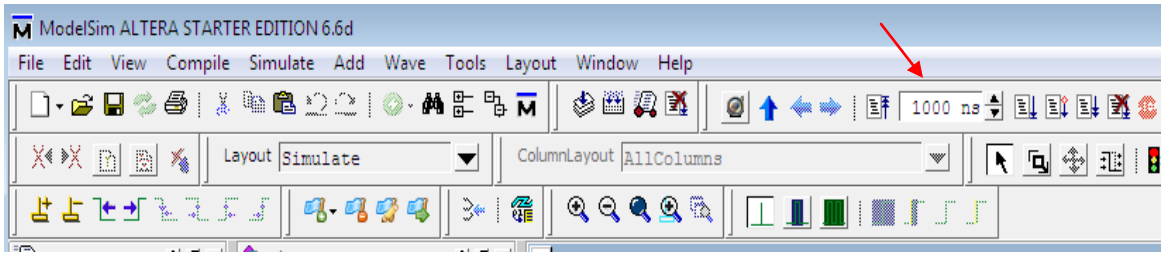
10. Repeat Steps 8 and 9 for the x signal in the Object pane. But use 40 ns counter period because x is the 2^1 order signal and y is the 2^0 order signal. In the next tutorial, you will have 2^2 and 2^3 order input signals. In this case, you would set the counter period to 80 ns (= 2^2 * 20 ns) and 160 ns (= 2^3 * 20 ns) for 2^2 and 2^3 input signals, respectively. At this point, the wave pane should look like the one below:



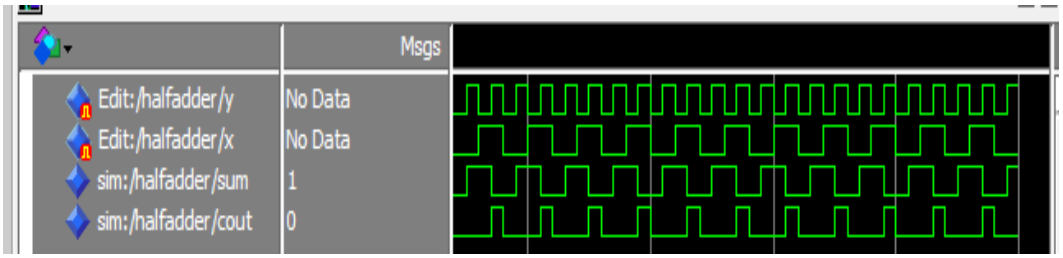
11. Right click on SUM signal in Object pane and Add → to Wave → Selected Signals. Repeat this step for the COUT signal. The wave pane should look like the one below:



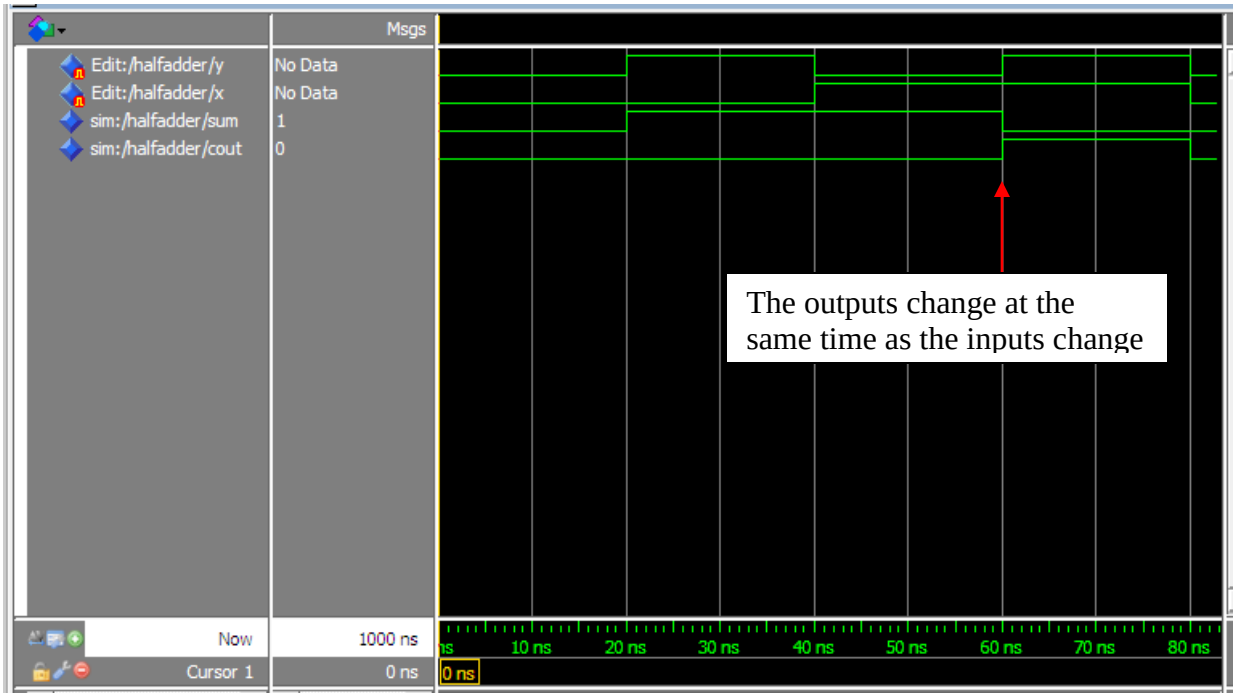
12. You can set the simulation end time in top menu bar. As shown, it is now set to 1000 ns. For the HalfAdder, you only need 80 ns simulation time to cover all possible outcomes for the four combinations of x and y input combinations.



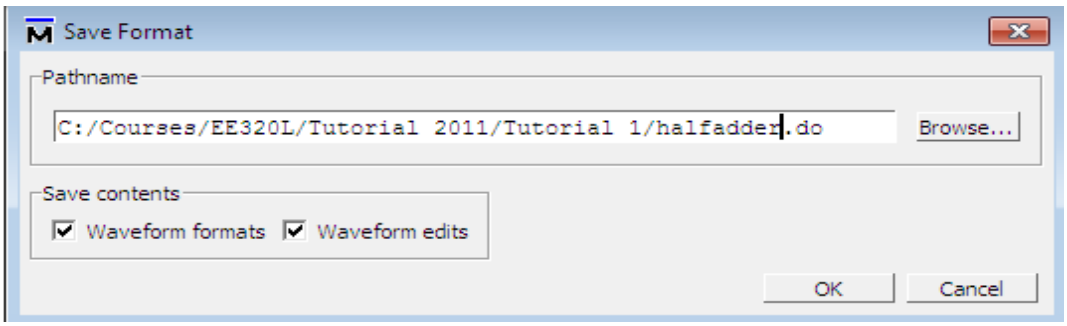
13. To run the simulation, Simulate → Run → Run-All. The simulation result is shown in the wave pane:



Zoom-in the wave window as many times as necessary so that you could convince yourself that there are no delay between input signals x and y, and the output signals SUM and COUT. Also, note that the simulation outputs are logically correct.



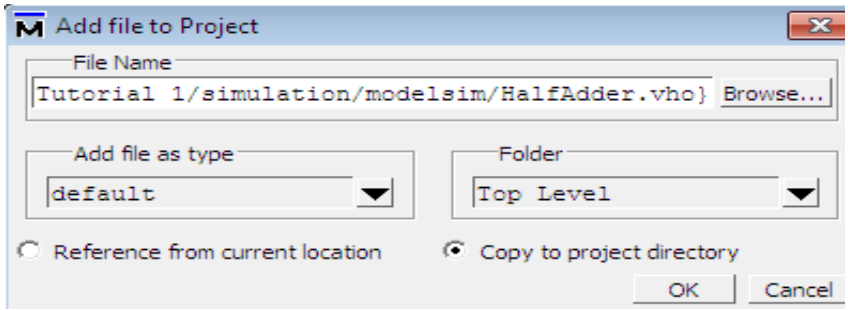
14. Save the simulation output and name it HalfAdder.do



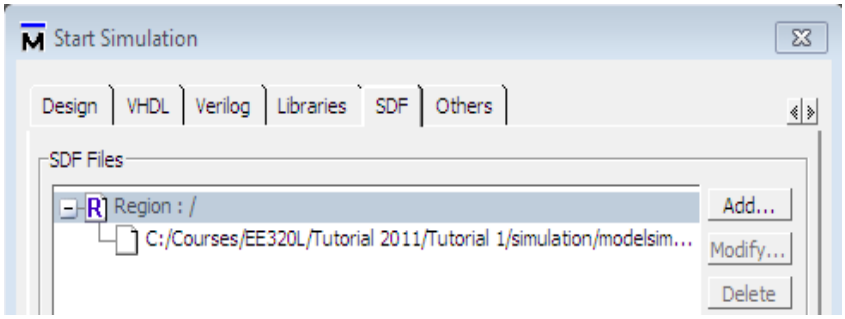
Simulate the half adder (Timing Simulation)

The difference in functional simulation and Timing simulation is that the Timing simulation needs halfadder.vho to supply the delay information. Quartus generates halfadder.vho and you need to include it in the project. The following steps will show you how to do it.

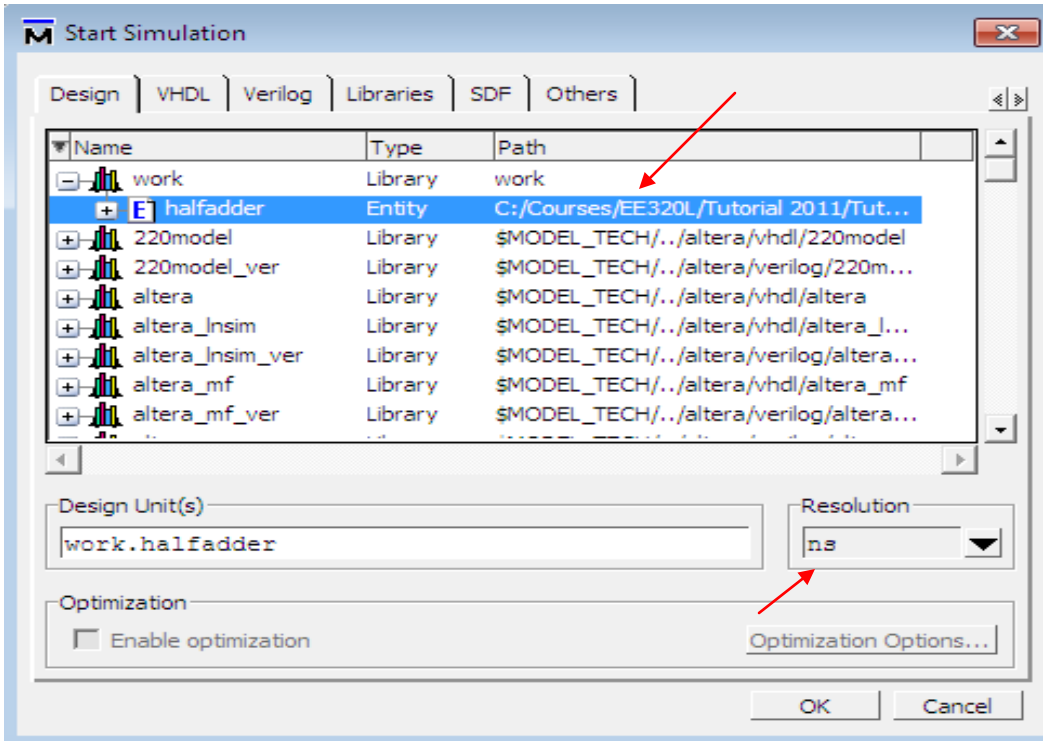
1. Open the project pane if it is not already open Right click anywhere in the project pane and Add to Project → Existing File. The Select a File to Add window appears. In this window, browse to simulation/Modelsim folder and select halfadder.vho. Mark Copy to project directory and click OK.



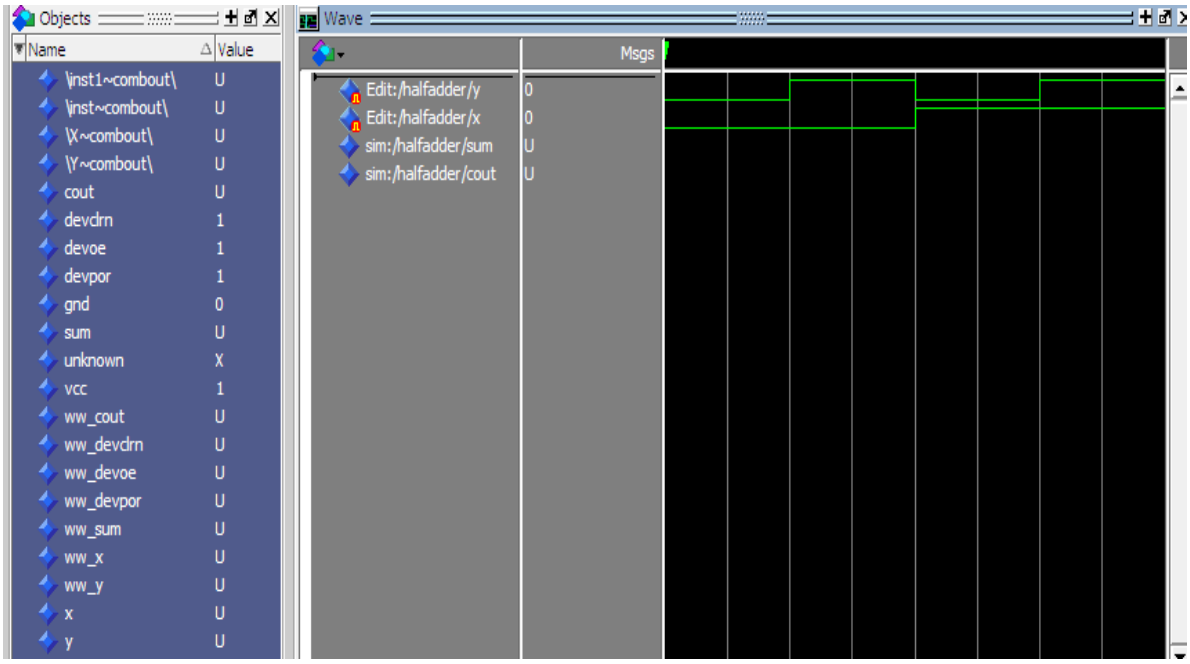
2. Right click anywhere in the project pane. Compile → Compile All. If halfadder.vho is not Top design, the compiler may give error messages. In this case, Compile → Compile Order. Click Auto Generate and click OK. Recompile all files: Compile → Compile All.
3. Simulate → Start Simulate. Start Simulation window appears. In this window, click SDF tab and then click add. Browse to simulation/Modelsim folder and select HalfAdder_vhd.sdo file.



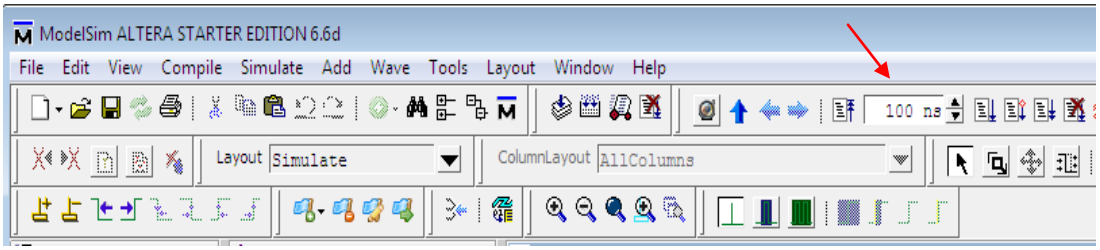
4. Click Design tab, expand work library and select HalfAdder_vhd.sdo file and select ns resolution. Click OK. SDF Backannotation Successfully Completed message should appear in the Transcript pane.



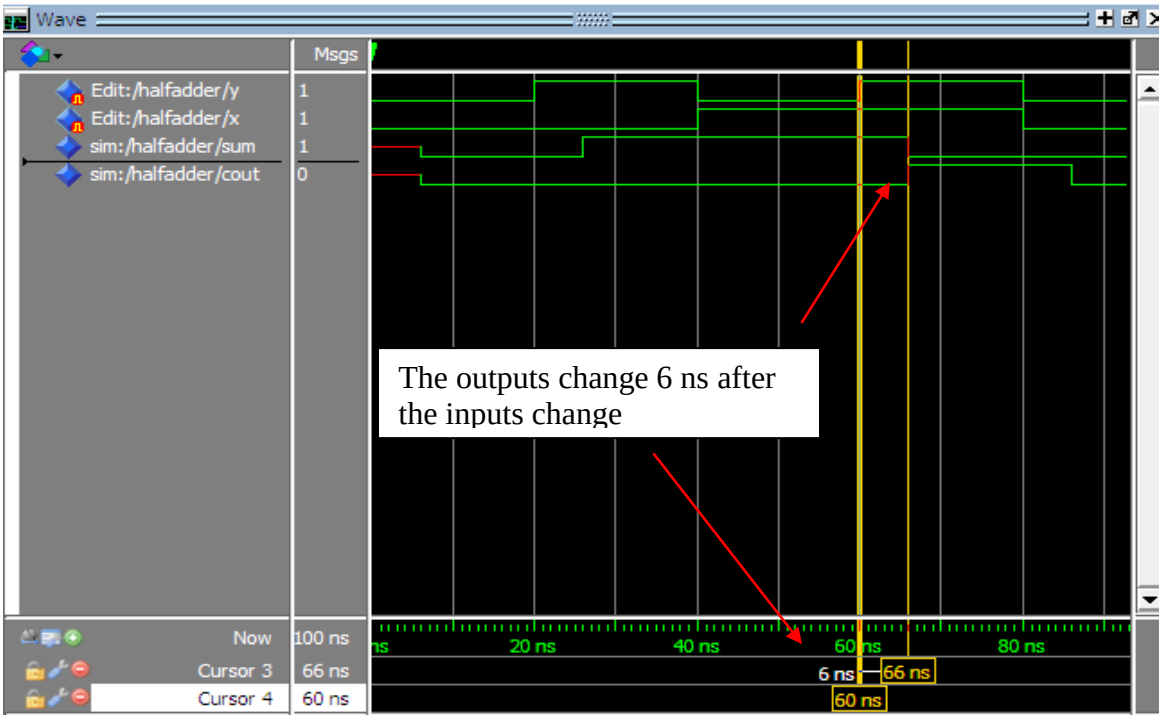
5. Follow Steps 7 to 11 in Functional Simulation section to set the input and output waveforms. If you do it correctly, wave pane should look like below. Another option is that you load the halfadder.do you saved for functional simulation. File → Load and select halfadder.do. The Load option, however, may load extra signals on the wave window and you may need to delete and re-arrange the signals.



6. You can set the simulation end time in top menu bar. Set it to 100 ns. For the HalfAdder, you only need 80 ns simulation time to cover all possible outcomes for the four combinations of x and y input combinations.



7. To run the simulation, Simulate → Run → Run-100. The simulation result is shown in the wave pane:



Note that there are 6 ns delay between input signals x and y, and the output signals SUM and COUT. Also, note that the simulation outputs are logically correct.

Note: When you want to re-simulate, reset the simulation first by Simulate → Run → Restart.

8. Save the simulation output and name it HalfAdderTiming.do